

WASP-35b

R-1725

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_141017 · created 2026-08-02T17:03:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456947.8625 +/- 0.0001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.023 +/- 0.047
Transit depth (Rp/Rs)^2	0.052 +/- 0.22 [%]
Orbital Inclination (inc)	83.28 +/- 2.0
Airmass coefficient 1 (a1)	0.65 +/- 0.2
Airmass coefficient 2 (a2)	0.15 +/- 0.051
Scatter in the residuals of the lightcurve fit is	26.72 %
Best Comparison Star	#1 - [171, 77]
Optimal Aperture	1.47
Optimal Annulus	12.21
Transit Duration (day)	0.046 +/- 0.043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.023 ± 0.047 vs 0.1241 (deviation 2.12σ)
Duration fit vs published	0.046 d vs 0.1317 d (deviation 1.97σ)
Mid-time O–C	0.1 min
Depth SNR	0.5
Residual scatter	26.72 %

Light curve

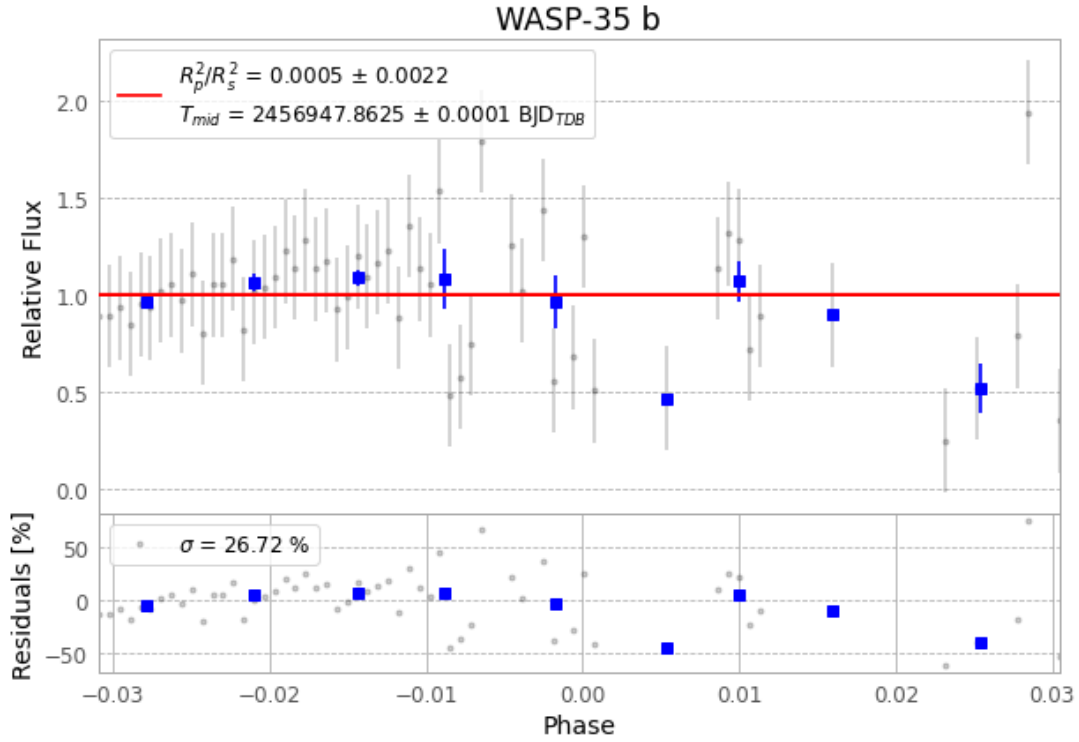


Figure 1 — FinalLightCurve_WASP-35 b_2014-10-16.png (SHA-256 0b4926f365a5ef7d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [180, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 add6893ed10a3f8e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

102 FITS frames (67 MB), WASP-35141017061512.FITS → WASP-35141017111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-141017.log (5709ed30bcfa...), FinalLightCurve_WASP-35 b_2014-10-16.png (0b4926f365a5...), FinalLightCurve_WASP-35 b_2014-10-16.csv (522afdb9d6ca...)

Compute resources

Wall time 290.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:03Z.